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Studierichting: Informatica
Oefeningenreeks 1D nr. 9.1

Bereken de waarde van $z^4 - z^3 - 5z^2 + 4z + 2$ voor $z = 1 + \sqrt{2}$:

$(1+\sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}$

$(1+\sqrt{2})^3 = 1 + 3 \cdot \sqrt{2} + 3 \cdot 2 + 2\sqrt{2} = 7 + 5\sqrt{2}$

$(1+\sqrt{2})^4 = (3+2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$

$\Rightarrow z^4 - z^3 - 5z^2 + 4z + 2 = (17+12\sqrt{2}) - (7+5\sqrt{2}) - 5(3+2\sqrt{2}) + 4(1+\sqrt{2}) + 2$

$= 17 + 12\sqrt{2} - 7 - 5\sqrt{2} - 15 - 10\sqrt{2} + 4 + 4\sqrt{2} + 2$

$= 1 + \sqrt{2}$

$$\boxed{z = 1 + \sqrt{2} \Rightarrow z^4 - z^3 - 5z^2 + 4z + 2 = 1 + \sqrt{2}}$$

References